

Quiz

Name : _____

November 5

1. A test for a disease called db is correct 95% of the time. That is, if someone has db, the test reports that they do 95% of the time, if someone does not have db, the test reports that they do not have db 95% of the time. The disease of db is rare, affecting 1 in ten thousand people.

What is the probability that an individual has db after a positive test (that is, a test that reports db is present)?

2. True or False?

- (a) Events A and A^c where $0 < P(A) < 1$ are never independent.
- (b) Two independent events are always also conditionally independent.
- (c) The Total Probability Theorem states that if $A_1, \dots, A_n$ are independent, $P(B) = P(A_1 \cap B) + P(A_2 \cap B) \dots + P(A_n \cap B)$
- (d) Events A and A^c partition the sample space.
- (e) If events Events A and B^c are independent, so are Events A^c and B^c